

Affect-Conditioned Decoding: Coupling a Lightweight Emotion Estimator to Sampling Parameters in Large Language Models

Omar Nuri

Independent Researcher

Computer Science and Machine Learning Systems

GitHub Repository:

github.com/Omar-S-Nuri/affect-conditioned-decoding

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Abstract

While autoregressive Large Language Models (LLMs) excel at generative tasks, their decoding pathways remain static and insensitive to shifting contextual affect or operational states. In this paper, we introduce a lightweight approach to Affect-Conditioned Decoding, where text inputs are mapped into a bounded 7D affect space via a multi-output regression matrix. The resulting vector is prepended to the input sequence while concurrently modulating downstream inference hyper-parameters (τ , top_k). We first expose a critical empirical hazard in evaluating such architectures: aggressive sampling restrictions alone induce localized mode collapse and repetitive token behavior in sub-billion parameter regimes (benchmarked on Qwen1.5-0.5B), which can be mistaken for genuine semantic alignment. To isolate the causal contribution of the affect vector from this sampling artifact, we design a rigorous 4-way ablation protocol combined with a double-blind human evaluation. Contrary to our initial hypothesis, we find *no* statistically significant evidence that the affect-vector prefix improves target-only perplexity, lexical diversity, or human-rated contextual appropriateness beyond hyperparameter throttling alone ($p > 0.05$ in all comparisons). We report this null result transparently and discuss a plausible explanation rooted in template overfitting during fine-tuning, arguing that it establishes a practical lower bound on the utility of lightweight, prefix-based affect conditioning for sub-billion parameter models.

1 Introduction

Autoregressive language transformers process data purely based on frozen, statistical word-embeddings and positional parameters. They possess no visceral

concept of biological urgency, self-preservation, or somatic continuity over multiple context windows. In contrast, biological organisms utilize a continuous, visceral state-evaluation loop to protect homeostasis.

By taking inspiration from neuroscientist Antonio Damasio’s *Descartes’ Error*, we investigate whether artificial agents can benefit from a lightweight, functional algorithmic analogue of continuous state-regulation, shifting inference from cold, disconnected token selection toward context-anchored, adaptive behavior. This framework models a two-stage system in which semantic prompt content alters an internal, continuously updated affect vector, which in turn constrains the downstream autoregressive token selection space.

Crucially, this work also evaluates the scaling limits of such affect-conditioned decoding, explicitly examining how a sub-billion parameter architecture reacts to severe inference constraints, and whether an affect-vector prefix meaningfully changes generation quality under those constraints or is simply overshadowed by them. By implementing an explicit multi-stage architecture that separates affect vectorization from token synthesis, and by evaluating it with a dedicated ablation protocol, this work aims to give an honest empirical account of where such interventions do, and do not, provide measurable benefit.

2 Related Work

Our approach interfaces directly with the established paradigms of controllable text generation, post-training steering, and inference-time parameter modulation in autoregressive transformer architectures.

2.1 Controllable Generation and Latent Steering

Controllable text generation has traditionally relied on dynamic weight modifications or auxiliary classifier gradients. A foundational baseline is the Plug and Play Language Model (PPLM) proposed by Dathathri et al., which steers text trajectories by computing gradients from a lightweight attribute model and back-propagating them into the transformer’s hidden states during the forward pass. While highly effective for complex tasks such as sentiment-steering and topic shifting, PPLM incurs substantial computational latency due to its iterative gradient updates at every generation step.

Subsequent frameworks explored post-training vocabulary modulation without altering base weights. Contrastive Decoding (CD), as formalized by Li et al., optimizes token distribution matrices by computing the log-prediction delta between a large expert model and an amateur counterpart. While CD effectively suppresses stylistic degenerations and hallucinations, it scales poorly due to the concurrent execution of multiple parameter-heavy networks.

2.2 Inference Layer and Sampling Parameter Optimization

Closer to the micro-architectural layer of token selection, dynamic sampling adjustments represent a highly efficient, gradient-free alternative for text steering. Classical methodologies, such as Top- k and Nucleus (Top- p) sampling introduced by Holtzman et al., dynamically truncate the long tail of the token probability distribution. This truncation reshapes the Softmax logit distribution to prevent degenerate repetitions or erratic, ungrammatical hallucinations.

Traditionally, however, these parameters (τ, k, p) remain either completely static throughout the entire inference cycle, or are conditioned strictly on the rolling token-probability entropy of the generated sequence itself.

Our work bridges the gap between these isolated paradigms. By introducing *Affect-Conditioned Decoding*, we avoid the latency of latent-space gradient back-propagation (as in PPLM) and the hardware overhead of dual-model architectures (as in Contrastive Decoding). Instead, our framework maps prompt semantics directly into a continuous, low-cost affect vector at the prompt frontier, conditioning both the sequence prefix and the inference constraints (τ, k) concurrently. As our results show, however, this low computational overhead does not by itself guarantee a measurable semantic benefit once hyperparameter throttling is already active – a distinction that, to our knowledge, prior work in this area does not explicitly isolate.

3 Methodology

Our architecture utilizes a dual-stage pipeline separating affect vectorization from language synthesis.

3.1 7D Affect Space Projection

A textual input stimulus X_t at time-step t is projected into a 7-dimensional affect space via a regularized multi-output Ridge regression matrix model $\mathcal{W}_{\text{PAL}}$ operating on top of character n-gram boundaries:

$$\mathbf{v}_t = \mathcal{W}_{\text{PAL}} \cdot \Phi(X_t) \quad (1)$$

where $\mathbf{v}_t = [V, A, D, DNG, RES, SOC, GOL]^T \in \mathbb{R}^7$ defines Valence, Arousal, Dominance, Danger (DNG), Resource Value (RES), Social Bond (SOC), and Goal Proximity (GOL). $\Phi(X_t)$ represents the sublinear TF-IDF character vectorizer profile ($2 \leq \text{n-gram} \leq 5$) protecting morphological edge cases. Each dimension is clipped to its own empirically observed training range rather than a shared $[0, 1]$ interval, since several dimensions (notably Valence) are naturally signed.

3.2 Sluggish Mood Dynamics and Affective Latency

The internal state $\mathbf{M}_t \in \mathbb{R}^7$ possesses algorithmic viscosity, bypassing discrete episodic retrieval in favor of a continuous, decay-based affect persistence loop. It tracks historical semantic shocks using an exponential smoothing protocol representing hormonal latency (e.g., adrenaline clearing rates):

$$\mathbf{M}_t = \alpha \cdot \mathbf{v}_t + (1 - \alpha) \cdot \mathbf{M}_{t-1} \quad (2)$$

where $\alpha = 0.3$ establishes the global clearing factor. Parallely, metabolic degradation is tracked by updating a scalar vitality gauge $E_t \in [0.05, 1.0]$ which tracks the computational tax of generating N tokens:

$$E_t = \max(0.05, E_{t-1} - (\gamma_{\text{base}} + \delta \cdot N)) \quad (3)$$

with baseline metabolic depletion $\gamma_{\text{base}} = 0.03$ and token cost coefficient $\delta = 0.002$. Energy deficits actively feed into the affect vector by expanding the core threat coordinate: $\mathbf{M}_t[DNG] \leftarrow \min(1.0, \mathbf{M}_t[DNG] + (1.0 - E_t) \cdot 0.15)$.

3.3 Mechanistic Hyperparameter Coupling

If the cumulative processed threat exceeds a critical threshold ($\mathbf{M}_t[DNG] > 0.6$), a behavioral freeze (tunnel vision) is triggered by reshaping the Softmax activation distribution:

$$\tau_t, k_t = \begin{cases} 0.25, 15 & \text{if } \mathbf{M}_t[DNG] > 0.6 \\ 0.40, 25 & \text{else if } E_t < 0.25 \\ 0.75, 50 & \text{otherwise} \end{cases} \quad (4)$$

where τ_t defines generation temperature and k_t represents the *Top-K* token filter. The final probability profile for token y_i is computed as:

$$P(y_i | y_{<i}, X_t, \mathbf{M}_t) = \text{Softmax} \left(\frac{z_i}{\tau_t} \right)_{k_t} \quad (5)$$

4 Experimental Evaluation and Quantitative Analysis

The quantitative validation of the proposed affect-conditioned decoding architecture was benchmarked across $N = 60$ balanced text stimuli, evenly split between benign and threatening scenarios. To isolate the causal influence of the 7D affect vector from sampling-induced artifacts, we designed a 4-way ablation study.

4.1 Ablation Design

The evaluation used a **Target-Only** perplexity methodology, in which prompt context tokens were fully masked out (*ignore_index* = -100) during cross-entropy loss computation, so that only the prediction difficulty of the generated continuation is measured – under a fixed, prefix-free scoring context shared across all four conditions. This decouples the measurement from the length or content of whatever prefix was used during generation, avoiding a confound present in naive full-prompt perplexity comparisons. Concurrently, the **Distinct-2** metric tracked bigram diversity, and a semantic embedding similarity score (*Sim*) was computed via a local **all-MiniLM-L6-v2** Sentence-Transformer against category-specific reference behaviors.

The four conditions were: (1) **absolute_control** (default sampling, no prefix), (2) **noise_padding_control** (default sampling, a semantically empty filler prefix of identical character length to the real affect vector), (3) **hyperparameter_baseline** (restrictive sampling, no prefix), and (4) **full_framework** (restrictive sampling, real affect-vector prefix). Conditions 2 and 3 exist specifically to separate two potential confounds – prefix length alone, and sampling restriction alone – from any genuine semantic contribution of the affect vector in Condition 4.

4.2 Testing for Causal Semantic Steering: A Null Result

Table 1 shows a pattern that must be interpreted carefully. Target-only perplexity is uniformly low across *all four* conditions (PPL between 1.06 and 1.21), including **1_absolute_control**, which uses neither a restrictive sampling regime nor any prefix. This is notably different from what naive, non-masked perplexity measurements on earlier iterations of this framework suggested, and indicates that at least part of the

previously observed perplexity collapse was an artifact of scoring the prefix together with the continuation rather than the continuation alone.

The **Distinct-2** diversity score does decline moderately from 0.546 (**1_absolute_control**) to 0.465 (**4_full_framework**), consistent with some loss of lexical variety under restrictive sampling. However, a directional Mann-Whitney-U test comparing **4_full_framework** against both **2_noise_padding_control** and **3_hyperparameter_baseline** on target-only perplexity found *no* statistically significant difference ($p > 0.05$ for both comparisons). In other words, once prefix length and sampling restriction are separately controlled for, we find no evidence that the semantic content of the affect vector itself causes a measurable reduction in perplexity beyond what an equally long, meaningless prefix or restrictive sampling alone already produce.

4.3 Double-Blind Human Evaluation

To test whether any subtler quality difference exists that the automatic metrics might miss, we conducted a double-blind human evaluation ($N = 30$ stratified prompt scenarios, drawn from the same 60-item pool). Using a dedicated preprocessing pipeline (**prepare_human_eval.py**), responses from **3_hyperparameter_baseline** and **4_full_framework** were anonymized under randomized unique identifiers (**ITEM_XXXXXX**) and presented to raters in shuffled order to prevent order bias.

Three independent human annotators, blind to condition, scored each response on a 1–5 Likert scale along two dimensions: *Syntactic Coherence* and *Contextual Appropriateness*. A paired, non-parametric Wilcoxon signed-rank test was used (*alternative* = 'greater', testing whether **4_full_framework** scores exceed **3_hyperparameter_baseline** scores).

Both conditions maintained a comparably high grammatical standard ($\emptyset = 4.10$ for **3_hyperparameter_baseline** vs. $\emptyset = 4.07$ for **4_full_framework**). Contextual appropriateness, however, was low in an absolute sense for both conditions, and the point estimate for **4_full_framework** ($\emptyset = 1.93$) was numerically *lower*, not higher, than for **3_hyperparameter_baseline** ($\emptyset = 2.13$). The paired Wilcoxon test confirmed that this difference was not statistically significant ($p = 0.9456$), and in particular provides no support for the hypothesis that the affect-vector prefix improves perceived contextual appropriateness over hyperparameter throttling alone.

Taken together, both the automatic ablation and the human evaluation converge on the same conclusion: under severe sampling restriction, the sub-billion parameter model used here does not exhibit a measurable, attributable benefit from the affect-vector prefix,

Table 1: Results of the 4-way ablation study: target-only perplexity (PPL), bigram diversity (Distinct-2), and semantic cosine similarity (Sim). Values are sample averages across $N = 60$ balanced evaluation contexts.

#	Evaluation Condition	Sampling Parameters	PPL ↓ ($\emptyset$)	PPL (σ)	Distinct-2 ↑	Sim ↑
1	1_absolute_control	$\tau = 0.75, k = 50$	1.21	0.35	0.546	0.387
2	2_noise_padding_control	$\tau = 0.75, k = 50$	1.06	0.08	0.455	0.382
3	3_hyperparameter_baseline	$\tau = 0.25, k = 15$	1.09	0.16	0.486	0.382
4	4_full_framework	$\tau = 0.25, k = 15$	1.07	0.13	0.465	0.380

Note: Statistical comparisons between Condition 4 and Conditions 2/3 were performed using a one-sided, non-parametric Mann-Whitney-U test (*alternative* = 'less') on the per-item Target-Only PPL distributions.

whether judged by target-only perplexity, lexical diversity, semantic similarity, or blinded human ratings.

5 Discussion and Limitations

The central empirical finding of this work is a null result, and we discuss it as such rather than reframing it. Two explanations seem most plausible, and are not mutually exclusive.

First, restrictive sampling ($\tau = 0.25, k = 15$) alone appears sufficient to drive the underlying Qwen1.5-0.5B model toward narrow, repetitive continuations (as reflected in the Distinct-2 decline in Table 1), largely independent of whatever prefix precedes the prompt. Under sufficiently tight token selection, there may simply be too little room left in the search space for a semantic prefix to meaningfully redirect generation.

Second, the fine-tuning procedure used to adapt the base model to the fixed "REIZ: ... REAKTION:" template (finetune_llm_verhaltensvielfalt.py, using a small number of training steps on a narrow template) may have caused the model to become highly confident about continuing this specific scaffold regardless of prefix content. The uniformly low target-only perplexity across *all four* conditions in Table 1, including the fully unconstrained 1_absolute_control, is consistent with this explanation: a model that has essentially memorized how to continue the fine-tuning template would show low perplexity on that continuation almost independent of what precedes it. We flag this as template overfitting rather than genuine cognitive alignment, and consider it the most likely reason the affect vector’s semantic content did not produce a measurable effect in either the automatic or the human evaluation.

We note this residual behavior for completeness: in informal, non-systematic spot checks, threat-related affect values were observed to persist for a few turns after an acute stimulus subsided (e.g., a modifier lingering into a subsequent benign prompt) before decaying under the smoothing rule in Eq. 2. We report this only as a qualitative observation about the decay mechanism, not as evidence of any generation-quality benefit, since our formal ablation and human evalua-

tion did not find one.

We consider a higher-capacity neocortex (e.g., $\geq 8B$ parameters) a necessary follow-up before drawing conclusions about whether affect-vector prefixes are inherently ineffective, or only ineffective under the specific combination of aggressive sampling restriction and a small, narrowly fine-tuned base model used in this study.

6 Conclusion and Future Work

In this work, we presented a computationally lightweight framework for Affect-Conditioned Decoding and, more importantly, a rigorous 4-way ablation and double-blind human evaluation protocol for testing whether such prefix-based semantic conditioning provides a measurable benefit once sampling-parameter effects are properly controlled for. Using this protocol on Qwen1.5-0.5B, we found no statistically significant evidence that the affect-vector prefix improves target-only perplexity, lexical diversity, semantic similarity, or human-rated contextual appropriateness beyond hyperparameter throttling alone. We consider this a meaningful negative result: it suggests a practical lower bound on the utility of this class of lightweight, prefix-based intervention in sub-billion parameter, narrowly fine-tuned models, and highlights template overfitting as a likely confound in evaluating such systems more broadly.

Future work should prioritize (1) repeating this ablation on larger, less narrowly fine-tuned base models (8B+ parameters) to test whether the null result is specific to small, template-overfit models; (2) reducing or removing the aggressive sampling restriction to test whether the affect vector has measurable effect under less severe throttling; and (3) expanding the evaluation set beyond $N = 60$ automatic and $N = 30$ human-rated items to increase statistical power, particularly given the modest effect sizes observed here.

AI Disclosure

The author utilized Claude (Anthropic) as an AI-assisted tool for formal phrasing of mathematical formulations and syntactic refinement of the manuscript,

including revising the framing of the results to accurately reflect the reported statistical outcomes. All empirical data collection, architectural framework design, algorithmic implementation, and intellectual ownership remain exclusively with the human author.

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